

Alyssa Wu

zjwualyssa@gmail.com || www.linkedin.com/in/zjwualyssa

Education

Pomona College — Computer Science, Public Policy Analysis (Science, Technology, Society) B.A

Claremont, CA

GPA: 3.98 / 4.0

Expected May 2028

Notable Courses: Human Centered Computing, Computational Design Tools, Computer Systems, Data Structures, Artificial & Biological Vision, Discrete Math & Functional Programming, Space & GIS, Linear Algebra, Vector Calculus, Differential Equations

Bergen County Academics — High School Diploma, International Baccalaureate Diploma

Hackensack, NJ

GPA: 3.99 / 4.0 | IB Diploma: 43/45 | Academy of Business and Finance

Class of 2024

Experience

The Hive (The Rick and Susan Sontag Center for Collaborative Creativity) — Data Analyst

Jan 2025 - Present

- Collaborate with makerspace staff and program teams to understand participant needs and engagement patterns, using behavioral data to inform event programming and outreach
- Build new attendance tracking system with automated Python/Excel pipelines integrating attendance and facility-usage data across Airtable, Eventbrite, and other touchpoints; present findings to donors, consortium presidents, and community stakeholders

Microsoft — Undergraduate Research Intern

May 2026 - Aug 2026

- Investigated county-level power outage prediction given a hurricane path using embeddings from Aurora, a foundational weather model, alongside socioeconomic and engineered storm features
- Constructed leakage-free forecast and reanalysis datasets across multiple timesteps and hurricanes to evaluate weather-model embeddings, storm features, and socioeconomic features across 5+ ML architectures; visualized predictions against ground truth across 10 hurricanes using custom GUI
- Developed MCP-based skills for a power system agent, integrating scheduled outages with traceable powerflow analysis, crew dispatch, and belief state updates to be visualized on a geospatial GUI

The Nature Conservancy — Artificial Intelligence Software Engineer Intern

May 2025 - July 2025

- Developed a SPECTER-based embedding model achieving 0.96 recall for relevance prediction of agroforestry literature based on title and abstract
- Conducted comparative experiments on 15+ text-embedding and neural architecture combinations using scikit-learn, PyTorch, and Hugging Face
- Built and delivered a deployable Google Colab for the Tackle Climate Change's (TCC's) team to streamline agroforestry evidence synthesis

Finpublica — Environmental Social Governance Finance Intern

Jan 2025 - Aug 2025

- Maintained membership and donation database, including the integration of 4000+ new members post-merger using Python and Excel
- Curated sustainable tech insights in agriculture and transportation for workshops and stakeholder reports

Newark Office of U.S. Senator Cory Booker — Intern

Sept 2023 - Jun 2024

- Attended to, classified, and tracked 100+ daily calls and voicemails to conduct constituent database cleaning and casework data visualization
- Represented Senator during site visits, with focus on Port Newark hybridization, \$16 billion Gateway Tunnel Project, and NY/NJ ferry subsidization

Projects

Reel Hunters — Computational Design Tool

Spring 2026

- Iteratively designed, prototyped, and built mobile experience with ReactNative exploring how short-form video platforms could support more intentional content consumption, inspired by the interaction patterns of Pokémon GO and Instagram Reels

Detecting Building Natural Disaster Vulnerability for American Red Cross

Fall 2025 - Spring 2026

- Collaborated with nonprofit to develop computer vision pipelines in Python and GIS to extract building characteristics and vulnerability indicators from 360° street-view imagery for resilience mapping; evaluated object detection approaches including YOLO, Faster R-CNN, and SpheroNet

P-AI Project Audio (Sampling Detection in Hip Hop with Deep Learning)

Fall 2025

- Investigated the robustness of audio fingerprinting and deep learning approaches for automatic music sample identification, examining how machine representations capture perceptual similarities in remixed music

Can I Get There Without A Car? (Claremont Consortium Accessibility Map)

Fall 2025

- Designed participatory ArcGIS accessibility map enabling students to document destinations, transportation modes, and barriers to car-free mobility; translated student-reported experiences into an interactive visualization to be presented to Claremont's Active Transportation Committee

Skills, Certifications, & Interests

Python | Java | C | PyTorch | TensorFlow | Figma | ReactNative | ArcGIS | Git | Excel | Bloomberg Market Concepts Certified | Bloomberg Environmental Social Governance Certified | Cantonese fluency | Mandarin proficiency | *Interests:* Earthen Urban Design, Classical Piano, Fabrication

Awards

Princeton Prize in Race Relations (2024) — Founded student-led BHM lesson initiative (20+ lesson plans, 100+ classes taught) & organized free tutoring in underrepresented towns to combat academic segregation as president of Bergen County Academics' Black and Latine Student Union